



MCS5210 – Control Engineering IV

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Introduction to Nonlinear Systems

- Nonlinearity is ubiquitous, and almost **all** systems are inherently **nonlinear**.
- Thus, all control systems are nonlinear to a **certain** extent.
- Nonlinear control systems can be described by **nonlinear differential equations**.
- However, if the operating range of a control system is small, and if the involved nonlinearities are smooth, then the control system may be reasonably **approximated** by a **linearized** system.
- Then significant amount of linear system analysis tools and control design tools can then be applied.
- However, there are **intrinsic** nonlinear behaviours which **cannot** be described using linear systems.
- Analysis and control of these systems are necessarily based on nonlinear systems.
- Even for a linear system, if there are **uncertainties**, nonlinear control strategies such as **adaptive** control may have to be used.



Nonlinear functions and nonlinearities

- Consider the system described by

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

- If the system matrices A, B, C and D are constant matrices, the system described by the state equation above is a linear time-invariant system.
- Nonlinear dynamic systems are the dynamic systems that contain at least **one nonlinear** component.
- That is, the functions in the differential equations **contain** nonlinear functions.
- A function $f: R \rightarrow R$ is linear if for any $x, y, k \in R$,

$$f(kx) = kf(x)$$

$$f(x + y) = f(x) + f(y) \Rightarrow \textit{superposition}$$

- A function that does **not** satisfy the above conditions is a **nonlinear** function.



Nonlinear functions and nonlinearities

- Consider the system below with input saturation

$$\dot{x} = Ax + B\sigma(u)$$

$$y = Cx + Du$$

Where $\sigma(u)$ is a **saturation** function defined as

$$f(x) = \begin{cases} -1, & u < 0 \\ u, & \text{otherwise} \\ 1, & u > 1 \end{cases}$$

- It is clear that the saturation function σ is a nonlinear function thereby making the system a nonlinear system.
- The superposition principle does not apply, because after the input saturation, any **increase** in the **input** amplitude does not lead to a **change** the system response .
- A general nonlinear system is often described by

$$\dot{x} = f(x, u)$$

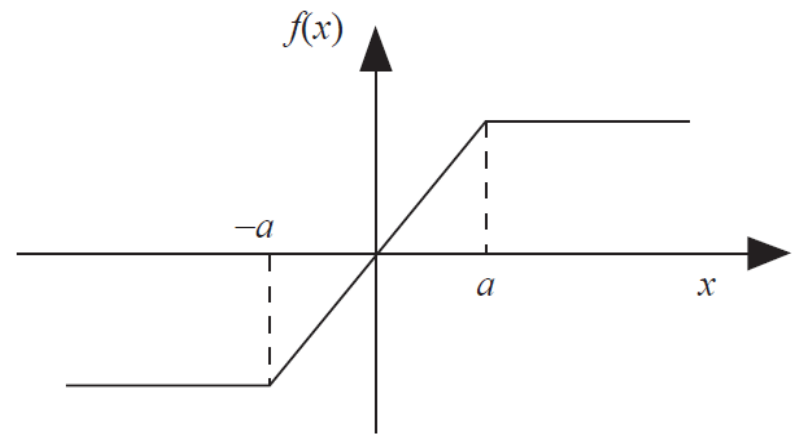
$$y = h(x, u)$$

f and h are smooth nonlinear functions



Nonlinear functions and nonlinearities

- Nonlinearities can also be classified in terms of their mathematical properties, as **continuous** and **discontinuous**.
- Discontinuous nonlinearities cannot be locally approximated by **linear** functions and are also called "**hard**" nonlinearities.
- Hard nonlinearities for instance, **saturation**, **hysteresis** and **dead zones** are commonly found in control systems.
- **Saturation**: when the input to a physical device is small, its increase leads to a corresponding (often proportional) increase in the output.
- However, when the input reaches a certain level, its further increase produces little or **no** increase in the output.
- The output simply stays at its **maximum** value and the device is said to be in **saturation** when this happens.

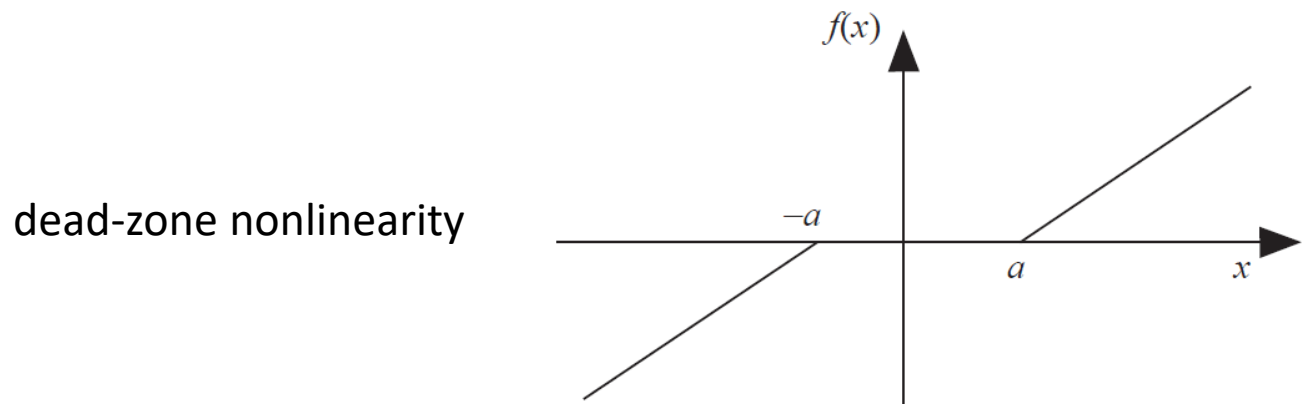


saturation function



Nonlinear functions and nonlinearities

- In many physical devices, the output is **zero** until the magnitude of the input exceeds a certain value.
- Such an input-output relation is called a **dead-zone**.
- Dead-zones can have a number of possible effects on control systems - the most common of which is to decrease static output **accuracy**.
- They may also lead to **limit cycles** or system **instability** because of the lack of response in the dead zone.
- In some cases, however, they may actually stabilize a system or **suppress** self-oscillations.
- For example, dead-zones could be employed in 2-step (**On/Off**) control systems to avoid **chattering** in the actuator.





Common Nonlinear System Behaviours

- Many nonlinear systems may be approximated by **linearised** systems around operating points.
- Their behaviours in close **neighbourhoods** can thus be predicted from the linear dynamics.
- This is the justification of applying linear system theory to control study, as almost all the practical systems are nonlinear systems.
- There are, however, many nonlinear **features** that do not exist in linear dynamic systems.
- Therefore, linearised dynamic models **cannot** describe the behaviours associated with these nonlinear features.



Common Nonlinear System Behaviours

- ✓ **Multiple equilibrium points:** These are common for nonlinear systems, unlike the linear system.
 - For example the system $\dot{x} = -x + x^2$ has two equilibria at $x = 0$ and $x = 1$.
 - This system's behaviours around these equilibrium points are very different, and they cannot be described by a **single** linearized model.
- ✓ **Limit cycles:** This a phenomenon that **periodic** solutions exist and attract nearby trajectories in positive or negative time.
 - Closed curve solutions may exist for linear systems, such as solution to harmonic oscillators.
 - But these are not attractive to nearby trajectories and not robust to any disturbances.
 - Heart beats of human body can be modelled as limit cycles of nonlinear systems.



Common Nonlinear System Behaviours

- ✓ **High-order harmonics and subharmonics:** These occur in the system output when subject to a harmonic input.
 - For linear systems, if the input is a harmonic function, the output is a harmonic function, with the same frequency, but different amplitude and phase.
 - For nonlinear systems, the output may even have harmonic functions with **fractional** frequency of the input or **multiples** of the input frequency.
 - This phenomenon is common in power distribution networks.
- ✓ **Finite time escape:** This can happen in a nonlinear system, i.e., the system state tends to infinity at a finite time.
 - This will never happen for linear systems.
 - Even for an unstable linear system, the system state can only grow at an exponential rate.



Common Nonlinear System Behaviours

- ✓ **Finite time convergence** to an equilibrium point can happen to nonlinear systems.
 - Nonlinear systems can be designed to achieve fast convergence.
 - This, again, cannot happen for linear systems, as the convergence rate can only be exponential.
- ✓ **Chaos** can only happen in nonlinear dynamic systems.
 - For some class of nonlinear systems, the trajectories are bounded, but do **not** converge to any equilibrium or limit cycles.
 - They may have quasi-periodic solutions, and the behaviour is very difficult to predict.



Autonomous Systems

- Consider the system

$$\dot{x} = f(x, u, t), \quad x(0) = x_0$$

$$y = h(x, u, t)$$

- For the system above, the external influence can only be exerted through the **input** u .
- If a system does not take any input, its future state only depends on the **initial** state.
- It means that the system is not influenced by external factors, and the behaviours are completely determined by the system state.
- Such a system is often referred to as an **autonomous** system.
- In terms of differential equations, an autonomous system can be expressed as

$$\dot{x} = f(x)$$

where $x \in R^n$ is the system state and $f: R^n \rightarrow R^n$



Autonomous Systems

- For an autonomous system $\dot{x} = f(x)$ a point x_e such that $f(x_e) = 0$ is a **singular** point.
- For an autonomous system, the set of all the trajectories provides a complete geometrical representation of the dynamic behaviour.
- This is often referred to as the **phase portrait** or phase plane.
- This is especially the case for second order systems in the format

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \phi(x_1, x_2)$$



Second Order Nonlinear System Behaviours

- A second order system may be written as

$$\dot{x}_1 = f(x_1, x_2)$$

$$\dot{x}_2 = f(x_1, x_2)$$

For an equilibrium point (x_{1e}, x_{2e}) , the linearised model is given by

$$\begin{bmatrix} \dot{\bar{x}}_1 \\ \dot{\bar{x}}_2 \end{bmatrix} = A_e \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix},$$

where

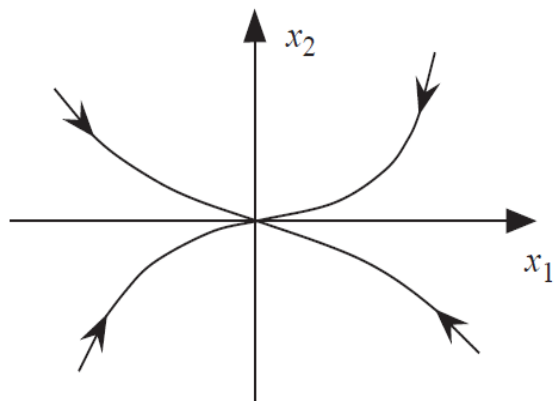
$$A_e = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(x_{1e}, x_{2e}) & \frac{\partial f_1}{\partial x_2}(x_{1e}, x_{2e}) \\ \frac{\partial f_2}{\partial x_1}(x_{1e}, x_{2e}) & \frac{\partial f_2}{\partial x_2}(x_{1e}, x_{2e}) \end{bmatrix}.$$

- The behaviour about this equilibrium or singular point is determined by the properties of matrix A_e .

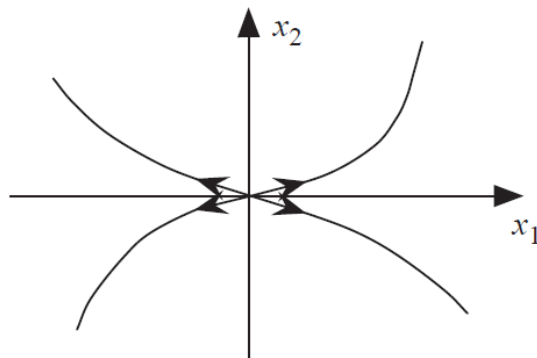


Classification of Singular Points

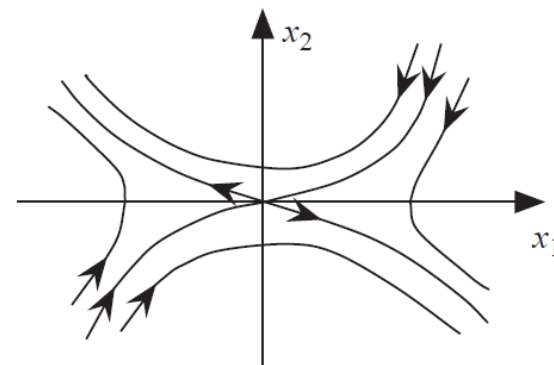
The singular points can be classified based on the **eigenvalues** λ_1 and λ_2 of the matrix A_e .



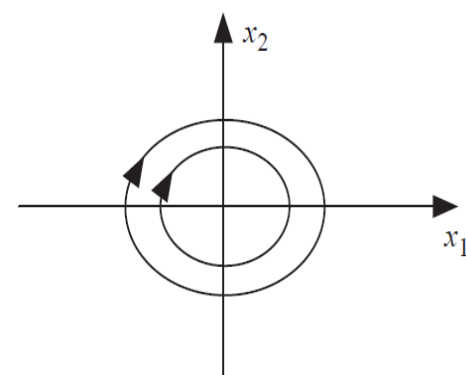
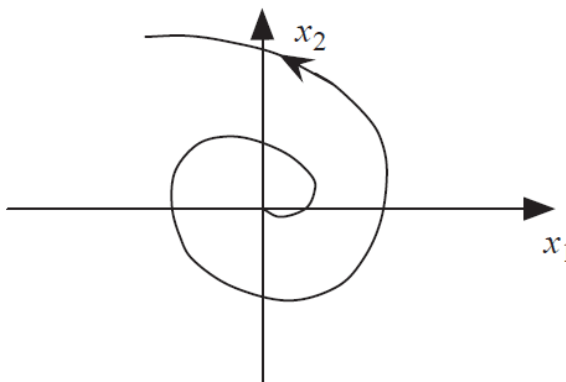
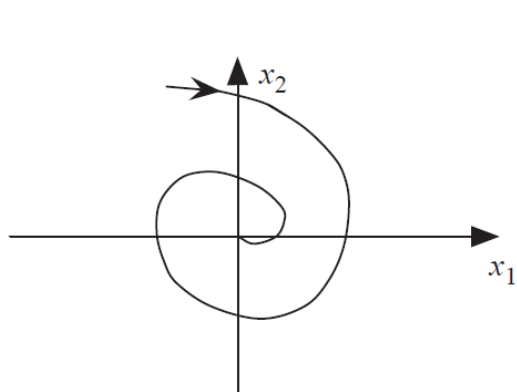
Stable node $\lambda_1 < 0, \lambda_2 < 0$



Unstable node $\lambda_1 > 0, \lambda_2 > 0$



Saddle point $\lambda_1 < 0, \lambda_2 > 0$



Stable focus $\lambda_{1,2} = \mu + jv, \mu < 0$ Unstable focus $\lambda_{1,2} = \mu + jv, \mu > 0$ Center $\lambda_{1,2} = \pm jv, v > 0$



Limit Cycles

- **Oscillation** is one of the most important phenomena that occur in dynamical systems.
- A system **oscillates** when it has a **periodic** solution in which some trajectories appear as closed curves in the phase portrait.
- For linear systems, periodic solutions appear in harmonic oscillators.
- For example a harmonic oscillator described by the system below has solutions as sinusoidal functions.

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1$$

- The solutions of this oscillator are **circles** around the **origin** and the amplitude is determined by the initial values.
- Also the solution for this harmonic oscillator is not robust, as any small disturbance will destroy the cycle.



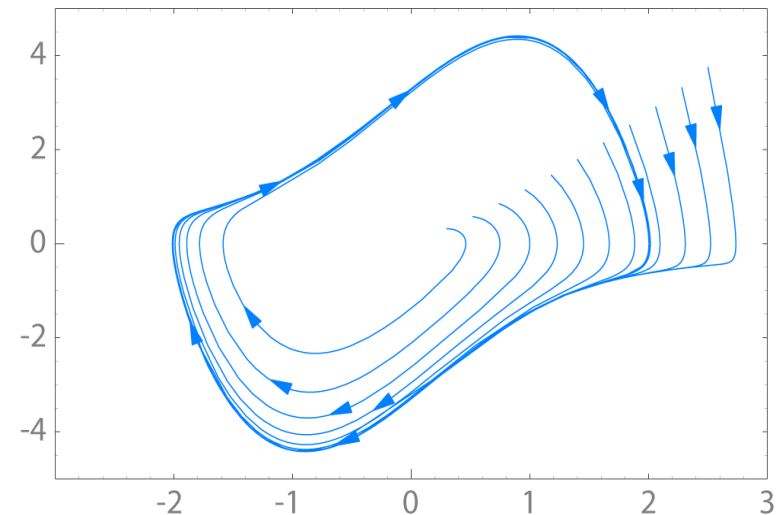
Limit Cycles

- A limit cycle is **stable** if nearby trajectories converge to it asymptotically, and unstable if they move away.
- One property of a limit cycle is that the amplitude of the oscillation may **not** depend on the initial values.
- A limit cycle may be **attractive** to the nearby region.
- One of the most famous ones is **van der Pol** oscillator.
- This oscillator can be viewed as a mathematical model of an RLC circuit, with the resistor being possible to take negative values in certain regions.
- One form of the van der Pol oscillator is described as
- By taking $x_1 = y$ and $x_2 = \dot{y}$ the system may be represented in state space as

$$\ddot{y} - \mu(1 - y^2)\dot{y} + y = 0, \quad \mu > 0$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 + \mu(1 - x_1^2)x_2$$





Poincare–Bendixson Theorem

- If a trajectory of the second-order autonomous system remains in a finite region, then **one** of the following is true:
 - ✓ *The trajectory goes to an equilibrium point.*
 - ✓ *The trajectory tends to an asymptotically stable limit cycle.*
 - ✓ *The trajectory is itself a cycle.*
- For high-order nonlinear systems, there are more complicated features if the trajectories remain in a bounded region which are beyond the limits of this course.



Lyapunov Stability Theory

- For control systems, design, one important objective is to ensure the **stability** of the closed-loop system.
- For a linear system, the stability can be evaluated in the time domain or frequency domain, by checking the **eigenvalues** of the system matrix or the **poles** of the transfer function.
- For nonlinear systems, the dynamics of the system **cannot** be described by equations in linear state space or transfer functions in general.
- Thus there is a need for more **general** definitions about the stability of nonlinear systems.
- More general concepts of stability are based on **Lyapunov** functions.

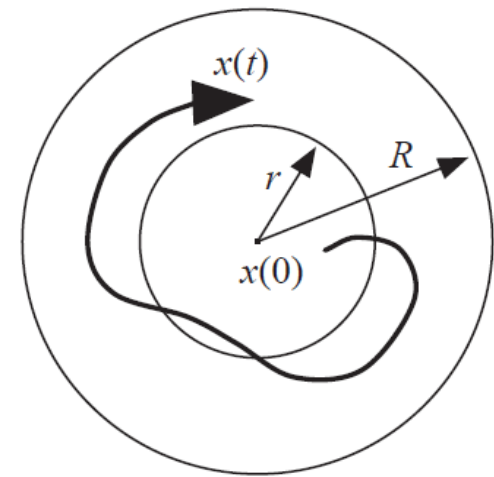


Lyapunov Stability: Definitions

- Consider the nonlinear system

$$\dot{x} = f(x)$$

- Lyapunov Stability**: the equilibrium point $x = 0$ is said to be **Lyapunov** stable if for any given positive real number R , there exists a positive real number r to ensure that $\|x(t)\| < R$ for all $t \geq 0$ if $\|x(0)\| < r$.
- Otherwise the equilibrium point is **unstable**.
- The definition of Lyapunov stability concerns with the behaviours of a dynamic system with respect to the **initial** state.
- If a system is Lyapunov stable, a restriction on the initial state of the system can be imposed to make sure that the state variables stay in a **certain** region.
- Critically** stable linear systems are **Lyapunov** stable.

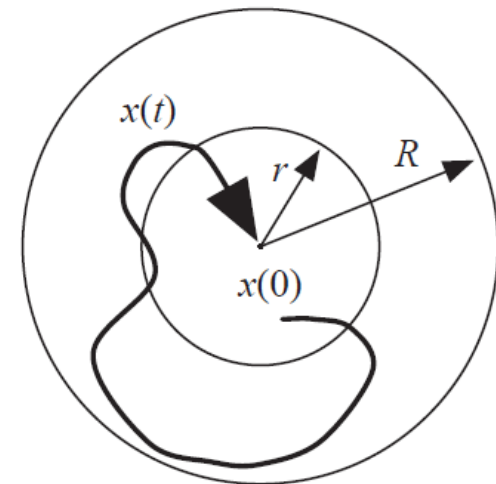


Lyapunov stability



Lyapunov Stability: Definitions

- **Asymptotic Stability:** An equilibrium point $x = 0$ is asymptotically stable if it is stable (Lyapunov) and if in addition there exists some $r > 0$ such that $\|x(0)\| < r$ implies that $\lim_{t \rightarrow \infty} x(t) = 0$.
- **Linear** systems with poles in the **left** side of complex plane are **asymptotically** stable.
- The asymptotic stability only requires that a solution **converges** to the equilibrium point.
- It does **not** however, specify the **rate** of convergence.



Asymptotic stability



Lyapunov Stability: Definitions

- **Exponential Stability:** An equilibrium point $x = 0$ is exponentially stable if there exist **two** positive real numbers α and λ such that the following inequality holds:

$$\|x(t)\| < \alpha \|x(0)\| e^{-\lambda t}$$

- If a **linear** system is asymptotically stable, it can be shown that it is **exponentially** stable.
- For nonlinear systems, a system may be asymptotically stable, but **not** exponentially stable.
- **Global Stability:** If asymptotic (exponential) stability holds for any initial state, the equilibrium point is said to be globally asymptotically (exponentially) stable (**GAS**).



Lyapunov's Direct Method

- **Positive Definite Functions:** A scalar function $V(x)$ is said to be locally positive definite if $V(0) = 0$ and in a ball $B_r, x \neq 0$ implies $V(x) > 0$. If the above properties hold for the entire space, the $V(x)$ is said to be globally positive definite.
- **Lyapunov Function:** If the function $V(x)$ is **positive** definite and has continuous partial derivatives, and if its time derivative is **negative** semidefinite, i.e $\dot{V}(x) \leq 0$, then $V(x)$ is a **Lyapunov** function.
- **Lyapunov Stability Theory:** If there exists a scalar function $V(x)$ with continuous first order derivatives such that
 - ✓ $V(x)$ is positive definite
 - ✓ $\dot{V}(x)$ is negative semidefinite

Then the equilibrium $x = 0$ point is stable.

- Furthermore, if $\dot{V}(x)$ is **negative** definite, then the stability is **asymptotic**.
- Stability analysis based on the Lyapunov function is the most **common** method to establish the stability of nonlinear dynamic systems.



Lyapunov's Direct Method

Example 1: Consider the nonlinear system

$$\dot{x} = -x^3, \quad x(0) = 0$$

Let

$$V = \frac{1}{2}x^2$$

Which is positive definite. Its derivative is given by

$$\dot{V} = -x^4$$

which is negative definite. Hence $x = 0$ is asymptotically stable.

Example 2: A pendulum can be described by

$$\ddot{y} + \dot{y} + \sin y = 0$$

Where y is the angle.

Let $x_1 = y$ and $x_2 = \dot{y}$. The dynamics can be represented in state space as

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\sin x_1 - x_2$$



Lyapunov's Direct Method

Consider the scalar function

$$V(x) = (1 - \cos x_1) + \frac{x_2^2}{2}$$

Which is a positive definite function for $|x_1| \leq \pi$, $x_2 \in R$.

Thus

$$\begin{aligned}\dot{V}(x) &= \sin x_1 \dot{x}_1 + x_2 \dot{x}_2 \\ &= -x_2^2\end{aligned}$$

Hence the system is stable at $x = 0$.

- The **drawback** of Lyapunov's direct method is that it is not always straight forward to **find** a Lyapunov function.
- In addition to analyzing system stability, Lyapunov functions are directly employed in some nonlinear control design methods to **guarantee** closed loop stability.